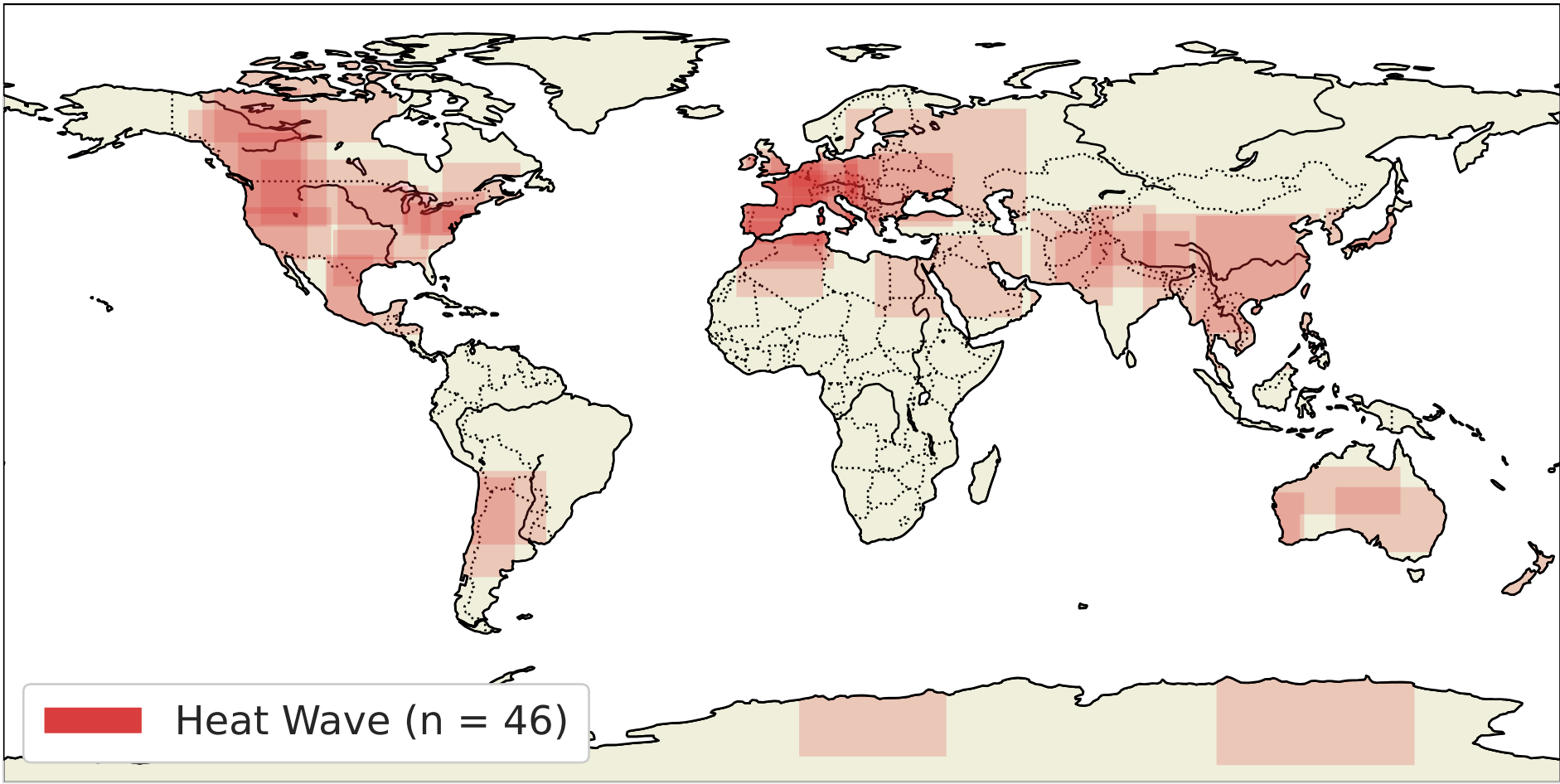
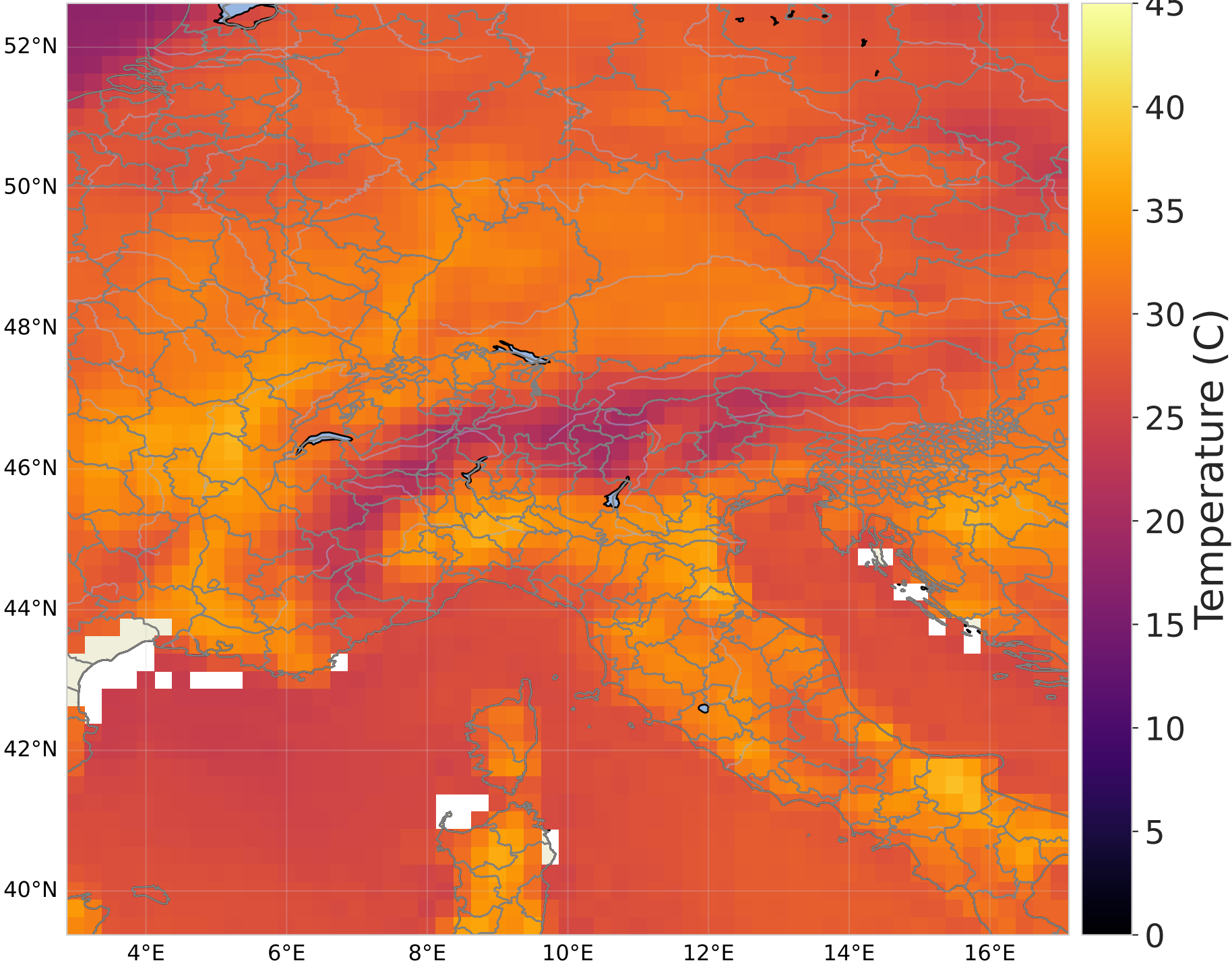


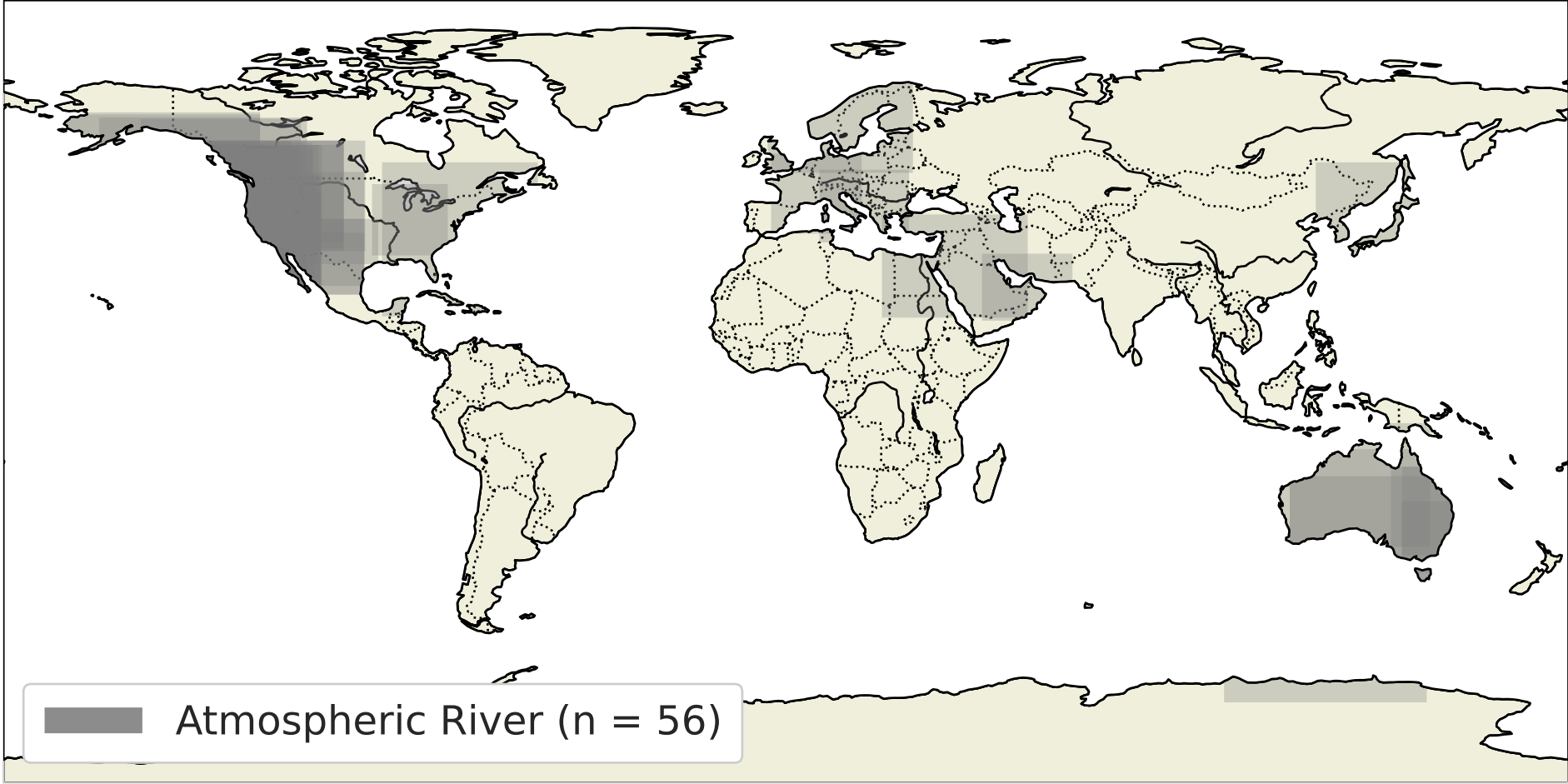
Heat Wave (n = 46)



Temperature Where > 85th Percentile Climatology
2023 Italy, Case ID 9
2023-07-11 12z



Atmospheric River (n = 56)



MAE of Event High

RMSE

MAE of Event Low

HRES IFS	1.9	2.1	2.1	2.2	4.9	1.8	2.1	2.6	3.2	4.2	1.4	1.4	1.5	1.8	2.8
FourCastNet v2	1.4	1.6	1.6	1.6	5.8	2.0	2.1	2.6	3.3	4.3	1.4	1.6	1.7	2.2	3.7
Panguweather	1.8	1.7	1.7	2.0	5.3	1.8	1.9	2.4	3.1	4.2	1.1	1.3	1.4	1.9	3.1
GraphCast	1.4	1.5	1.6	1.9	5.7	2.1	2.4	2.8	3.4	4.4	1.1	1.3	1.4	1.8	3.4
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
	Lead Time [Days]					Lead Time [Days]					Lead Time [Days]				

MAE of Event High

RMSE

MAE of Event Low

HRES IFS	0.7	1.1	0.3	0.5	6.0	1.6	1.8	2.2	2.8	3.6	1.5	1.1	1.5	2.2	1.7
FourCastNet v2	0.2	0.7	2.4	4.0	4.2	1.7	2.1	2.5	3.2	3.4	1.5	1.6	2.3	4.3	1.7
Panguweather	0.8	0.3	0.7	3.1	7.0	1.4	1.7	2.1	2.9	3.9	1.4	1.7	2.7	4.0	2.4
GraphCast	0.2	0.6	1.4	3.6	5.0	1.5	1.7	2.1	2.8	3.9	1.0	1.4	2.1	3.0	3.6
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
	Lead Time [Days]					Lead Time [Days]					Lead Time [Days]				

Critical Success Index

Spatial Displacement

HRES IFS	0.27	0.20	0.13	0.08	0.04	103	162	275	451	594
FourCastNet v2	0.31	0.23	0.13	0.07	0.03	110	194	277	447	625
Panguweather	0.33	0.25	0.16	0.09	0.04	101	195	285	430	582
GraphCast	0.46	0.32	0.21	0.12	0.05	95	173	267	431	631
	1	3	5	7	10	1	3	5	7	10
	Lead Time [Days]					Lead Time [Days]				

